PHYB57 F2024 - Introduction to Scientific Computing

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E-mail communication (preferred to Quercus messages)

Please send all e-mail correspondences from your university account and include your student number. In the email subject, include the prefix "PHYB57: "

Text Book

We will use the textbook "Python Programming And Numerical Methods: A Guide For Engineers And Scientists". There is no need to purchase it as there is a freely-available online version that will be extensively used:

https://pythonnumericalmethods.studentorg.berkeley.edu/notebooks/Index.html

Office hours:

Monday 1-2pm - default time is 5 mins per student

There will be regularly scheduled office hours during which the instructor will be available to engage in individual or group consultations as determined by the needs of the students. Should additional office hours be required, students are encouraged to make such requests, specifying a few preferred time slots that would be convenient for them.

Learning outcomes

- Develop proficiency in Python for numerical modeling and data analysis.
- Understand computational models and their applications in physics.
- Gain skills in scientific visualization and the interpretation of data.

Lectures

In-person lectures will be on Mondays 10-11am and Thursdays 11am - 12noon.

Note that the lectures will start 10 minutes past the hour. Please make sure you are on time.

Please refrain from eating lunch during the lectures.

Practicals

Mondays, 3-4pm or 4-5pm (students will be assigned to one of two 1hr-long sessions)

Attendance is mandatory for all practicals. During the practicals, exercises assigned to all students as practice will be solved/reviewed.

Grading Scheme

Assessment Title	Percent (%)
Homework (x4)	10
Pop quizzes (lectures & practicals)	10
Term test 1	20
Term test 2	20
Final Exam	40
Total Percentage	**100**

Academic Integrity in PHYB57

> Students in this course are obligated to understand what constitutes plagiarism. UTSC demands high standards of integrity and ethical conduct. We will follow a strict adherence to the University's Code of Behaviour for Academic Matters. You can find it in https://www.utsc.utoronto.ca/vpdean/academic-integrity-matters

There are also certain specific guidelines that apply to this course. While collaboration with fellow students is encouraged, it is prohibited to present any work that is not one's own original production. In the event of working with a partner, it is mandatory to indicate this on all assignments submitted. Failure to do so will be considered as a violation of academic integrity. It is important to note that even when working in partnership, each individual is required to produce their own distinct document and submitting identical documents will also be considered as a violation.

Furthermore, it is a violation of academic integrity to solicit the assistance of friends, family members or any other third party in completing assignments. Similarly, the hiring of external entities to produce assignments on one's behalf is strictly prohibited and shall be treated as a severe violation of academic integrity.

As a general rule, the use of the internet for research purposes is permissible, however, the sharing or distribution of any course-related materials such as lectures, assignments, quizzes or exams is strictly prohibited. This includes posting such material on public spaces such as discussion forums, and sharing it in private spaces such as instant messaging apps or chat groups. Any such violations will be treated as a severe breach of academic integrity.

The use of AI tools will be permitted, as detailed during the first lecture.

Accessibility

This course welcomes and accommodates students with a diverse range of learning styles and needs. If you have any disability or health condition that may require accommodations, please do not hesitate to contact the instructor or the AccessAbility Services Office for assistance. Your enquiries will be treated with the utmost confidentiality.

Problem Sets:

Handed out in class approximately every other week (total of 4). Due one week later.

Policy on collaboration: You are welcome to discuss the problems with fellow students, but you must write your own solutions, individually.

Policy on late problem set returns: In order to be fair to those who turn assignments in on time, points will be deducted on assignments turned in late.

Tentative Lecture Schedule:

Week 1

Introduction and overview. Chapter 1 - Python Basics

Week 2

Chapter 2 & 3 - Variables, Data Structures, Functions

Week 3

Chapter 4 & 5 - Branching & Iteration

Week 4

Chapter 8 & 9 - Complexity & Representation of Numbers

Week 5

Chapter 11 & 12 - Reading, Writing & Visualization

Week 6

Chapter 16 - Least Square Regression

Reading Week - No lecture or tutorial on February 23rd

Week 7

Chapter 17 - Interpolation

Week 8

Chapter 19 - Root Finding

Week 9

Chapter 20 - Numerical Differentiation

Week 10

Chapter 21 - Numerical Integration

Week 11

Chapter 22 - ODE IVPs

Week 12

Chapter 23 - - ODE BVPs